Universidad Politécnica Salesiana del Ecuador

Proyecto 10: Cadena de Hoteles "Gran Horizonte" (EIGRP + Aislamiento de Huéspedes)

Asignatura:

Redes de Computadora II

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Introducción

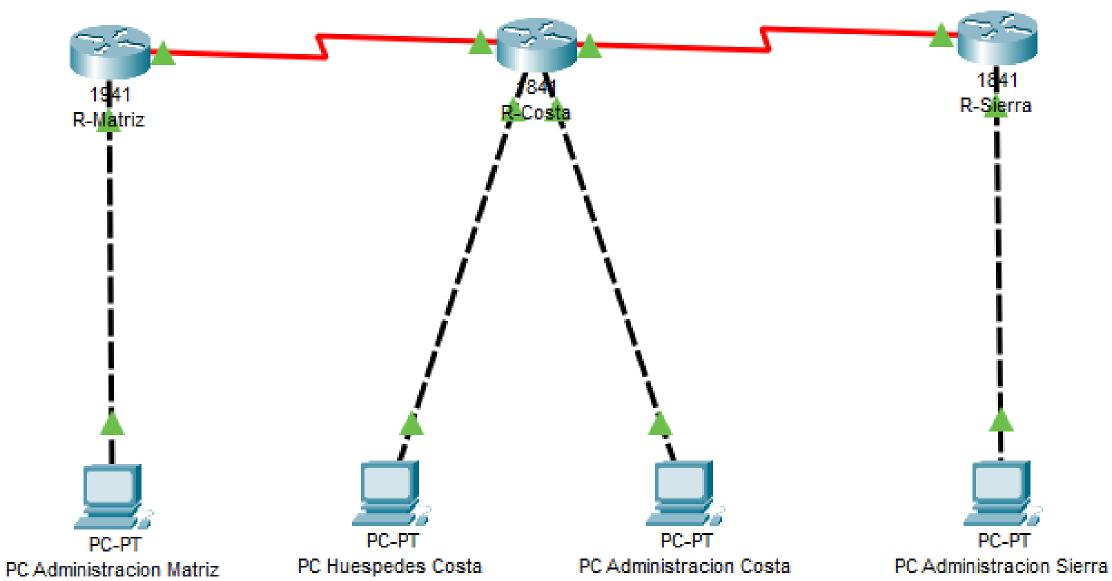
En este proyecto se realizó la interconexión de tres sedes (Matriz, Costa y Sierra) mediante enlaces WAN seriales. Se configuró direccionamiento IP para cada red y se implementó el protocolo de enrutamiento dinámico EIGRP para permitir la comunicación entre todas las sedes.

Además, se aplicaron listas de control de acceso (ACL) extendidas con el objetivo de restringir el acceso de las redes de huéspedes hacia los segmentos administrativos, garantizando así mayor seguridad y segmentación en la red. Finalmente, se realizaron pruebas de conectividad para verificar el correcto funcionamiento de la configuración.

Información de Direccionamiento

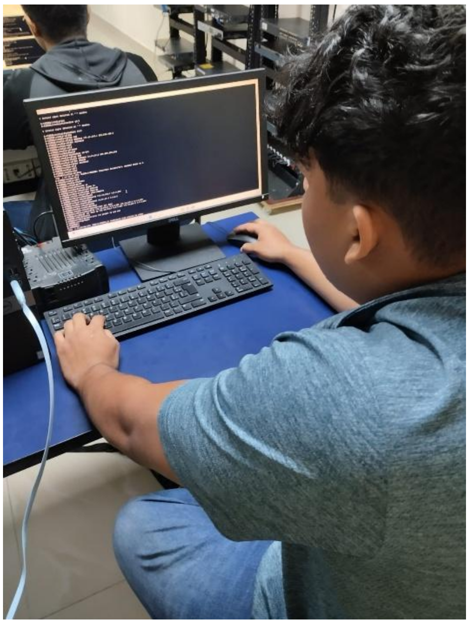
Dispositivo	Interfaz	Red	Máscara	Rango IP Útiles	Gateway
Matriz	G0/0	172.16.100.0	255.255.255.0	172.16.100.1 – 172.16.100.254	172.16.100.1
Costa	F0/0	172.16.101.0	255.255.255.0	172.16.101.1 – 172.16.101.254	172.16.101.1
Costa	F0/1	172.16.102.0	255.255.255.0	172.16.102.1 – 172.16.102.254	172.16.102.1
Sierra	F0/0	172.16.103.0	255.255.255.0	172.16.103.1 – 172.16.103.254	172.16.103.1
Matriz–Costa	S0/0/0	10.10.10.0	255.255.255.252	10.10.10.1 – 10.10.10.2	N/A
Costa–Sierra	S0/0/1	10.10.10.4	255.255.255.252	10.10.10.5 – 10.10.10.6	N/A

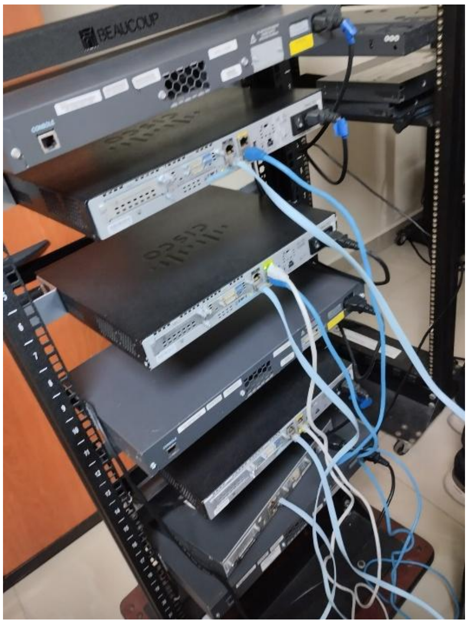
Topología

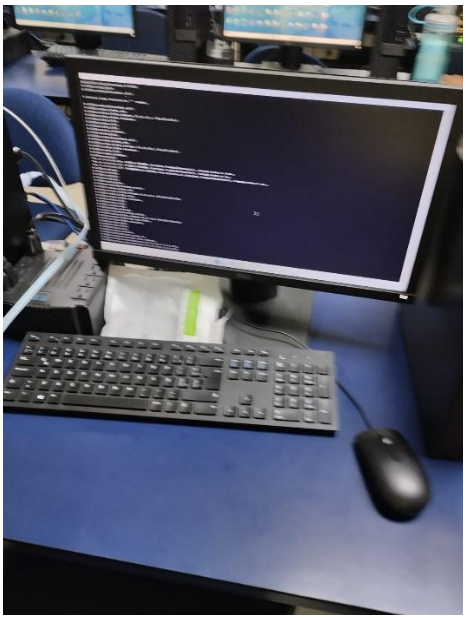


FOTOGRAFÍAS (Equipos reales)

En esta sección se presentan evidencias fotográficas de los equipos físicos utilizados en el laboratorio. Se muestran los routers interconectados mediante cables seriales para la comunicación WAN y cables Ethernet para las redes LAN. En las imágenes se puede observar claramente la conexión de los cables en los puertos correspondientes (Serial y FastEthernet), verificando la correcta instalación física de la topología implementada.







CONFIGURACIONES (Show Running-Config)

Se presentan las configuraciones completas de los routers implementados en la topología.

R-MATRIZ

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R-MATRIZ
R-MATRIZ(config)#int g0/0
                       ^
% Invalid input detected at '^' marker.

R-MATRIZ(config)#int fa0/0
R-MATRIZ(config-if)#ip address 172.16.100.1 255.255.255.0
R-MATRIZ(config-if)#no shutdown
R-MATRIZ(config-if)#
*Jan  1 00:24:32.675: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R-MATRIZ(config-if)#exit
R-MATRIZ(config)#int s0/0/0
R-MATRIZ(config-if)#io address 10.10.10.1 255.255.255.252
                     ^
% Invalid input detected at '^' marker.

R-MATRIZ(config-if)#ip address 10.10.10.1 255.255.255.252
R-MATRIZ(config-if)#clock rate 64000
R-MATRIZ(config-if)#no shutdown
R-MATRIZ(config-if)#
*Jan  1 00:26:05.119: %LINK-3-UPDOWN: Interface Serial0/0/0, changed state to do
wn
R-MATRIZ(config-if)#exit
R-MATRIZ(config)#router eigrp 10
R-MATRIZ(config-router)#no auto-summary
R-MATRIZ(config-router)#network 172.16.100.0
R-MATRIZ(config-router)#network 10.10.10.0
R-MATRIZ(config-router)#exit
R-MATRIZ(config)#ip route 0.0.0.0 0.0.0.0 200.1.1.2
R-MATRIZ(config)#router eigrp 10
R-MATRIZ(config-router)#redistribute static
R-MATRIZ(config-router)#
```

```
R-MATRIZ#show running-config
Building configuration...

Current configuration : 843 bytes
!
version 12.4
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R-MATRIZ
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
ip cef
!
!
!
!
!
!
!
!
!
!
interface FastEthernet0/0
 ip address 172.16.100.1 255.255.255.0
 duplex auto
 speed auto
!
interface FastEthernet0/1
 no ip address
 shutdown
 duplex auto
 speed auto
!
interface Serial0/0/0
 ip address 10.10.10.1 255.255.255.252
 clock rate 64000
!
interface Serial0/0/1
 no ip address
 shutdown
 clock rate 125000
!
router eigrp 10
 redistribute static
 network 10.0.0.0
```

```
!
!
!
!
!
!
interface FastEthernet0/0
 ip address 172.16.100.1 255.255.255.0
 duplex auto
 speed auto
!
interface FastEthernet0/1
 no ip address
 shutdown
 duplex auto
 speed auto
!
interface Serial0/0/0
 ip address 10.10.10.1 255.255.255.252
 clock rate 64000
!
interface Serial0/0/1
 no ip address
 shutdown
 clock rate 125000
!
router eigrp 10
 redistribute static
 network 10.0.0.0
 network 172.16.0.0
 no auto-summary
!
ip forward-protocol nd
ip route 0.0.0.0 0.0.0.0 200.1.1.2
!
no ip http server
no ip http secure-server
!
!
control-plane
!
!
line con 0
line aux 0
line vty 0 4
 login
!
scheduler allocate 20000 1000
end
```

R-COSTA

```
R-Costa#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R-Costa(config)#interface f0/0
R-Costa(config-if)#ip address 172-16.101.1 255.255.255.0
                                 ^
% Invalid input detected at '^' marker.

R-Costa(config-if)#ip address 172.16.101.1 255.255.255.0
R-Costa(config-if)#no shutdown
R-Costa(config-if)#exit
R-Costa(config)#interface f0/1
R-Costa(config-if)#ip address 172.16.102.1 255.255.255.0
R-Costa(config-if)#no shutdown
R-Costa(config-if)#exit
R-Costa(config)#interface s0/0/1
R-Costa(config-if)#ip address 10.10.10.5 255.255.255.252
R-Costa(config-if)#clock rate 64000
R-Costa(config-if)#no shutdown
R-Costa(config-if)#exit
R-Costa(config)#router eigrp 10
R-Costa(config-router)#no auto-summary
R-Costa(config-router)#network 172.16.101.0 0.0.0.255
R-Costa(config-router)#network 172.16.102.0 0.0.0.255
R-Costa(config-router)#network 10.10.10.0 0.0.0.3
R-Costa(config-router)#network 10.10.10.4 0.0.0.3
R-Costa(config-router)#$ip 172.16.101.0 0.0.0.255 172.16.100.0 0.0.0.255
R-Costa(config)#$ 100 deny ip 172.16.101.0 0.0.0.255 172.16.102.0 0.0.0.255
R-Costa(config)#access-list 100 permit ip any any
R-Costa(config)#interface f0/0
R-Costa(config-if)#ip access-group 100 in
R-Costa(config-if)#
```

```
R-Costa#show running-config
Building configuration...

Current configuration : 1171 bytes
!
version 12.4
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R-Costa
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
ip cef
!
!
!
!
!
!
!
!
!
!
interface FastEthernet0/0
 ip address 172.16.101.1 255.255.255.0
 ip access-group 100 in
 duplex auto
 speed auto
!
interface FastEthernet0/1
 ip address 172.16.102.1 255.255.255.0
 duplex auto
 speed auto
!
interface Serial0/0/0
 ip address 10.10.10.2 255.255.255.252
 clock rate 2000000
!
interface Serial0/0/1
 ip address 10.10.10.5 255.255.255.252
 clock rate 64000
!
router eigrp 10
 network 10.10.10.0 0.0.0.3
 network 10.10.10.4 0.0.0.3
 network 172.16.101.0 0.0.0.255
```

```
!
ip forward-protocol nd
!
no ip http server
no ip http secure-server
!
access-list 100 deny   ip 172.19.101.0 0.0.0.255 172.16.100.0 0.0.0.255
access-list 100 deny   ip 172.16.101.0 0.0.0.255 172.16.100.0 0.0.0.255
access-list 100 deny   ip 172.16.101.0 0.0.0.255 172.16.102.0 0.0.0.255
access-list 100 permit ip any any
!
control-plane
!
!
line con 0
line aux 0
line vty 0 4
 login
!
scheduler allocate 20000 1000
end
```

R-SIERRA

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R-SIERRA
R-SIERRA(config)#int fa0/0
R-SIERRA(config-if)#ip address 172.16.103.1 255.255.255.0
R-SIERRA(config-if)#no shutdown
R-SIERRA(config-if)#exit
R-SIERRA(config)#int
*Jan  1 00:23:22.687: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
% Incomplete command.

R-SIERRA(config)#int s0/0/0
R-SIERRA(config-if)#ip address 10.10.10.6 255.255.255.252
R-SIERRA(config-if)#no shutdown
R-SIERRA(config-if)#exit
*Jan  1 00:24:06.403: %LINK-3-UPDOWN: Interface Serial0/0/0, changed state to down
R-SIERRA(config)#router eigrp 10
R-SIERRA(config-router)#no auto-summary
R-SIERRA(config-router)#network 172.16.103.0 0.0.0.255
R-SIERRA(config-router)#network 10.10.10.4 0.0.0.3
R-SIERRA(config-router)#$ 101 deny ip 172.16.103.0 0.0.0.255 172.16.
```

```
                        ^
% Invalid input detected at '^' marker.

R-SIERRA(config-router)#$ip 172.16.103.0 0.0.0.255 172.16.100.0 0.0.0.255
R-SIERRA(config)#$ 101 deny ip 172.16.103.0 0.0.0.255 172.16.102.0 0.0.0.255
R-SIERRA(config)#access-list 101 permit ip any any
R-SIERRA(config)#int fa0/0
R-SIERRA(config-if)#ip access-group 101 in
R-SIERRA(config-if)#
```

’

```
R-SIERRA>enable
R-SIERRA#terminal length 0
R-SIERRA#show running-config
Building configuration...

Current configuration : 1014 bytes
!
version 12.4
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R-SIERRA
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
ip cef
!
!
!
!
!
!
!
!
!
!
interface FastEthernet0/0
 ip address 172.16.103.1 255.255.255.0
 ip access-group 101 in
 duplex auto
 speed auto
!
interface FastEthernet0/1
 no ip address
 shutdown
 duplex auto
 speed auto
!
interface Serial0/0/0
 ip address 10.10.10.6 255.255.255.252
 clock rate 2000000
!
interface Serial0/0/1
 no ip address
 shutdown
 clock rate 2000000
```

```
!
!
!
!
interface FastEthernet0/0
 ip address 172.16.103.1 255.255.255.0
 ip access-group 101 in
 duplex auto
 speed auto
!
interface FastEthernet0/1
 no ip address
 shutdown
 duplex auto
 speed auto
!
interface Serial0/0/0
 ip address 10.10.10.6 255.255.255.252
 clock rate 2000000
!
interface Serial0/0/1
 no ip address
 shutdown
 clock rate 2000000
!
router eigrp 10
 network 10.10.10.4 0.0.0.3
 network 172.16.103.0 0.0.0.255
 no auto-summary
!
ip forward-protocol nd
!
no ip http server
no ip http secure-server
!
access-list 101 deny   ip 172.16.103.0 0.0.0.255 172.16.100.0 0.0.0.255
access-list 101 deny   ip 172.16.103.0 0.0.0.255 172.16.102.0 0.0.0.255
access-list 101 permit ip any any
!
control-plane
!
!
line con 0
line aux 0
line vty 0 4
 login
!
scheduler allocate 20000 1000
end
```

PRUEBAS DE CONECTIVIDAD

Vecinos EIGRP

- **R-MATRIZ**

```
R-Matriz#show ip eigrp neighbors
IP-EIGRP neighbors for process 10
H   Address          Interface       Hold Uptime     SRTT    RTO   Q    Seq
                                     (sec)           (ms)          Cnt  Num
0   10.10.10.2       Se0/0/0         14    00:07:54  40      1000  0    8
```

- **R-COSTA**

```
R-Costa>show ip eigrp neighbors
IP-EIGRP neighbors for process 10
H   Address          Interface       Hold Uptime     SRTT    RTO   Q    Seq
                                     (sec)           (ms)          Cnt  Num
0   10.10.10.1       Se0/0/0         10    00:02:51  40      1000  0    27
1   10.10.10.6       Se0/0/1         12    00:02:51  40      1000  0    27
```

- **R-SIERRA**

```
R-Sierra>show ip eigrp neighbors
IP-EIGRP neighbors for process 10
H   Address          Interface       Hold Uptime     SRTT    RTO   Q    Seq
                                     (sec)           (ms)          Cnt  Num
0   10.10.10.5       Se0/0/0         14    00:08:11  40      1000  0    25
```

Tabla de Enrutamiento

- **R-MATRIZ**

```
R-MATRIZ#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     172.16.0.0/24 is subnetted, 1 subnets
C       172.16.100.0 is directly connected, FastEthernet0/0
```

- **R-COSTA**

```
R-Costa>show ip route

Codes: C - connected, S - static, I - IGRP, R - RIP,  M - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - INEF area
       N - OSPF NSSA external type 1, N2 - OSPF NSSA extenal type 2
       E - OSPF  external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS level-1, L2 - IS-IS level-2, ia - GCNG-PRR
       * - candidate default, U - per-user static route, o - ODR

Gateway of last reesort is not set

C     10.0.0./0 is subnetted, 2 subnets.
C     10.10.10.0 is directly connected, Seial0/0/0.
C     10.16.10.0.10 [90/2172416] via 10.10.10.0.2; 09, Seria0/0/1.
D     10.119.100.00 [90/2172416] via 10.10.10.10 00:2-19, Serial0/0/]
D     172.16.160.0/24 is directly connected, FastEthernet0/0
D     172.16.103.0 [90/2172416] viia 10.10.10.6, 00:29:49, Serial0/01
```

- **R-SIERRA**

```
R-Sierra>#how ip route

Codes: C - connected, S -  static, I - IGRP, R - RIP, M - mdbile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - Caal-6 inter area
       N - OSPF; SEA - EIGRP external 0   = Daal=8 NSA Extecteb-e type 2
       E - OSPFA NSSA external type 1, E1, 2- OSPF NSE external type 2
       B - IS-IS,,i1 - IS-IS saear=2  AIGPs 0 =
       u - periodic downloaded static route
Gateway of last proces is not set

D    10.0.0.7/7 is variabbly sudnetted, 3 subnets, 2 masfk.
D    10.16.10.8 [90/2172416] via 10.10.10.10, 00:05:22, Seriala10/0/0,
D    172.16.0.0 [90/307456)) via 10.10.10.10, 00:05:22, Serial0//0
D    192.168.10.0 [90/28810] via 10.10.10.10, 00:05:22, Serial00/0,
C    10.10.10.0.0 (30 isl irectly connected, Seria10/0/0/0
C    192.168.0.0/24S directly connected, FastEthenet0/0
```

ACL Aplicada

- **R-COSTA**

```
R-Costa#show access-lists
Extended IP access list 100
    10 deny ip 172.19.101.0 0.0.0.255 172.16.100.0 0.0.0.255
    20 deny ip 172.16.101.0 0.0.0.255 172.16.100.0 0.0.0.255
    30 deny ip 172.16.101.0 0.0.0.255 172.16.102.0 0.0.0.255
    40 permit ip any any
```

- **R-SIERRA**

```
R-SIERRA#show access-lists
Extended IP access list 101
    10 deny ip 172.16.103.0 0.0.0.255 172.16.100.0 0.0.0.255
    20 deny ip 172.16.103.0 0.0.0.255 172.16.102.0 0.0.0.255
    30 permit ip any any
```

Prueba de Ping

Ping que funciona (Administración <-> Administración)

- **ping 172.16.100.1**

```
Pinging 172.16.100.1 with 32 bytes of data:

Reply from 172.16.100.1: bytes=32 time<1ms TTL=255
Reply from 172.16.100.1: bytes=32 time<1ms TTL=255
Reply from 172.16.100.1: bytes=32 time<1ms TTL=255
Reply from 172.16.100.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.16.100.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping bloqueado (Huéspedes -> Administración)

- **ping 172.16.100.1**

```
Pinging 172.16.100.1 with 32 bytes of data:

Reply from 172.16.101.1: Destination host unreachable.
Reply from 172.16.101.1: Destination host unreachable.
Reply from 172.16.101.1: Destination host unreachable.
Reply from 172.16.101.1: Destination host unreachable.

Ping statistics for 172.16.100.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```